EFY TESTED
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WEATHER STATION With DEVICE CONTROL Using Touch-Screen Display

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This weather station provides such information as temperature and humidity at a place on a colourful touch-screen display. It can also control the temperature through the touch-screen user interface when it exceeds the set threshold level. The project can be easily converted to a personal weather station by adding more sensors to measure weather conditions outdoors or indoors.

The digital console provides easy readouts of the data collected by the sensor. The touch-screen has a microSD card memory slot. So, it can be connected to a personal computer where data can be displayed, stored, and uploaded to websites, or used for data ingestion or distribution systems. The author's prototype is shown in Fig. 1.

Components required for this project are mentioned in the Table.

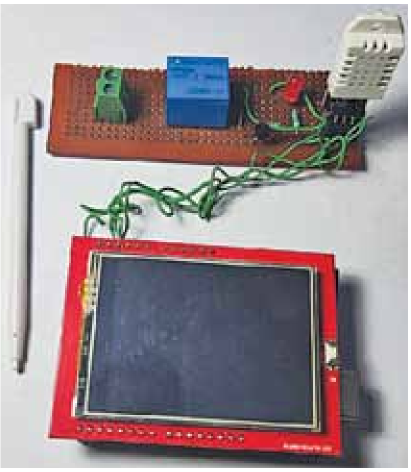

Fig. 1: Author's prototype

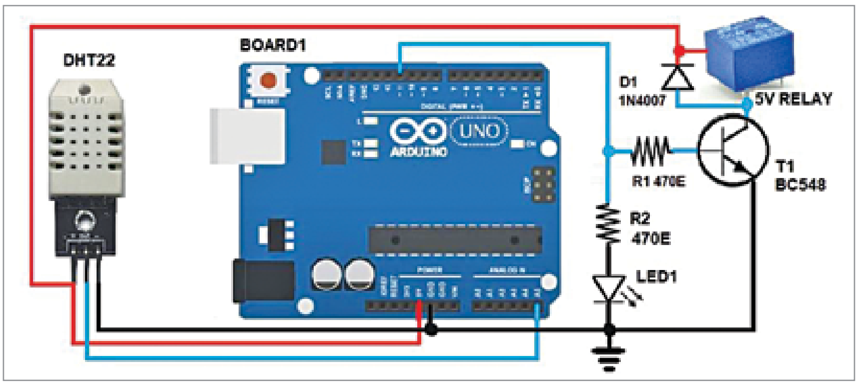

Fig. 2: Circuit diagram

COMPONENTS REQUIRED

Components	Quantity	Purpose
Arduino Uno	1	Computing
6cm TFT LCD touch screen shield	1	Data display
DHT22	1	Temperature and humidity sensor
BC548	1	NPN transistor amplifier for driving relay
Resistor (470-ohm)	2	Current limiter
1N4007	1	Rectifier diode used as freewheeling diode
Relay	1	5V SPDT relay for switching the device
LED	1	5mm LED for relay status indicator
Jumper wires	9	Connections
PCB	1	10x3cm Veroboard

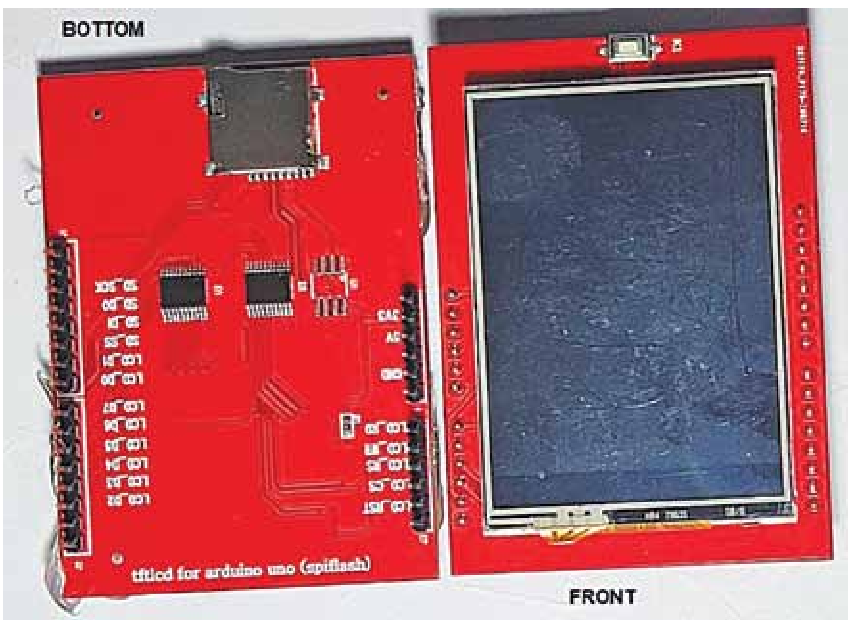

Fig. 3: TFT touch-screen display